A set is computable exactly when it and its complement are computably enumerable

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-computability-computability-theory/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is complete in-span: the forward direction is one composition and the converse is the h construction with the final membership equivalence

Editorial note. The label thm:ce-comp names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Kleene T-predicate. T(e,x,s) is a primitive recursive relation holding when s codes a halting computation of machine e on input x; phi_e(x) converges iff some s satisfies T(e,x,s).. Both halves rest on it: enumerability is witnessed by T-predicate search, and the minimal-s construction h(x) makes the characteristic function computable.

Theorem 0.1. *Let A be any set of natural numbers. Then A is computable if and only if both A and $\overline{A}$ are computably enumerable.*

Proof. The forwards direction is easy: if A is computable, then $\overline{A}$ is computable as well ($\chi A = 1 \dot{-} \chi\overline{A}$), and so both are computably enumerable.

In the other direction, suppose A and $\overline{A}$ are both computably enumerable. Let A be the domain of φ_d, and let $\overline{A}$ be the domain of φ_e. Define h by

$$h(x) = \mu s(T(d, x, s) \vee T(e, x, s)).$$

In other words, on input x, h searches for either a halting computation of φ_d or a halting computation of φ_e. Now, if $x \in A$, it will succeed in the first case, and if $x \in \overline{A}$, it will succeed in the second case. So, h is a total computable function. But now we have that for every x, $x \in A$ if and only if $T(e, x, h(x))$, i.e., if φ_e is the one that is defined. Since $T(e, x, h(x))$ is a computable relation, A is computable. □